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下一代大模型架构

Transformer的竞争者/后继者

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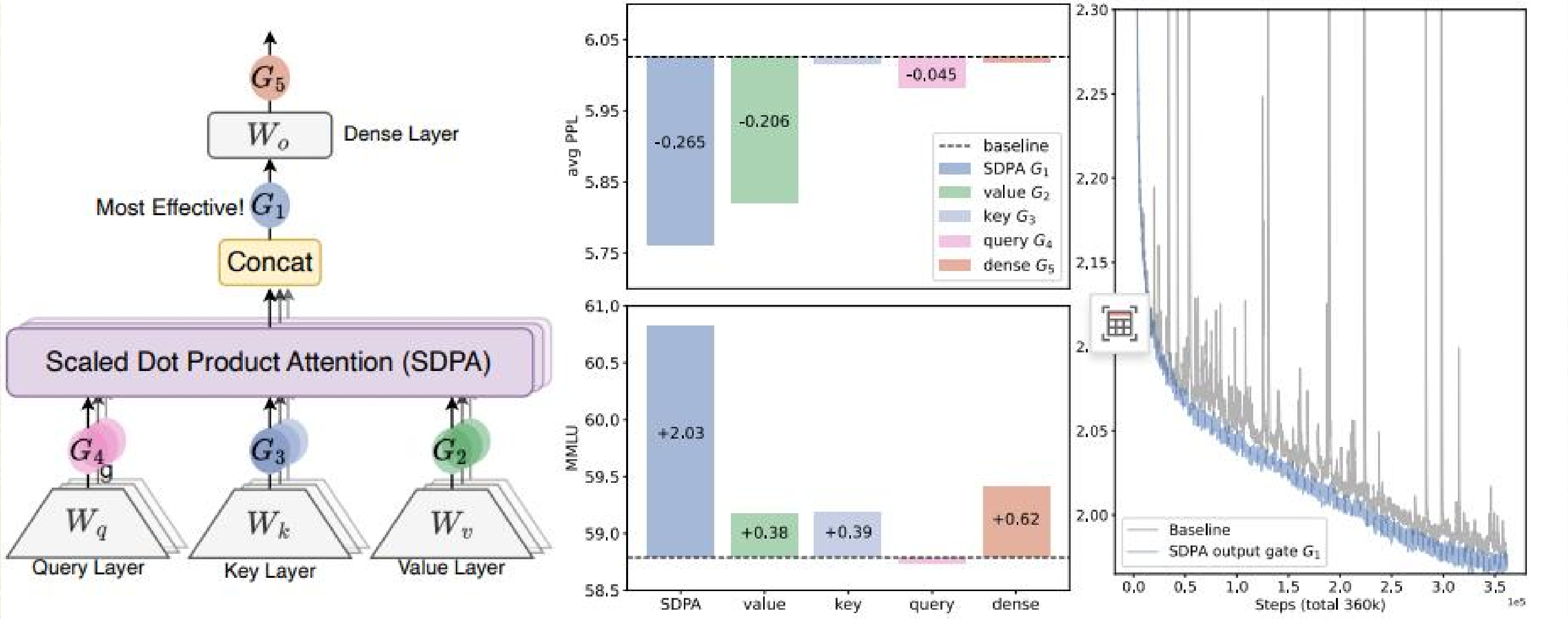
2025 NeurIPS best paper：Gated Attention

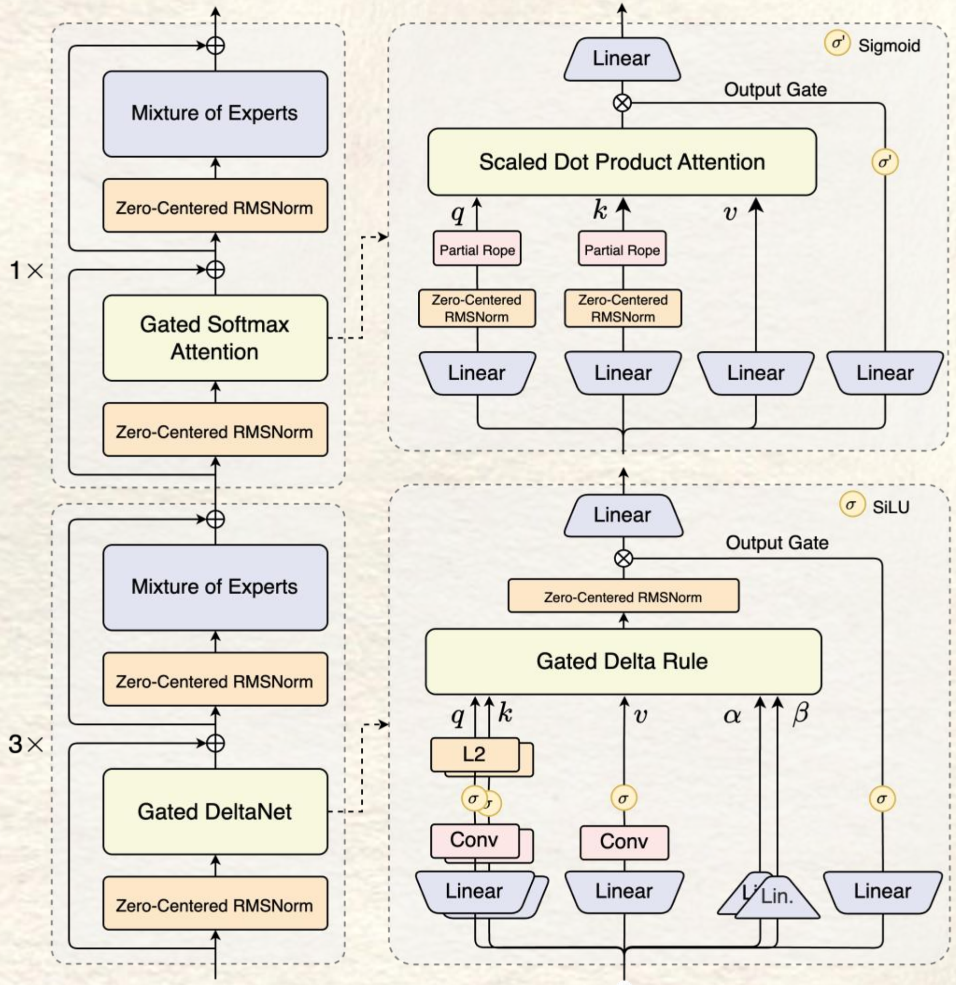


这不是今天重点，但它提出了一个问题：
更优秀的注意力架构到底长什么样？

Qwen-Next中25%用到了Gated Attention

https://arxiv.org/pdf/2505.06708

为什么 Qwen-next 有 75% 模块都是 Gated DeltaNet？

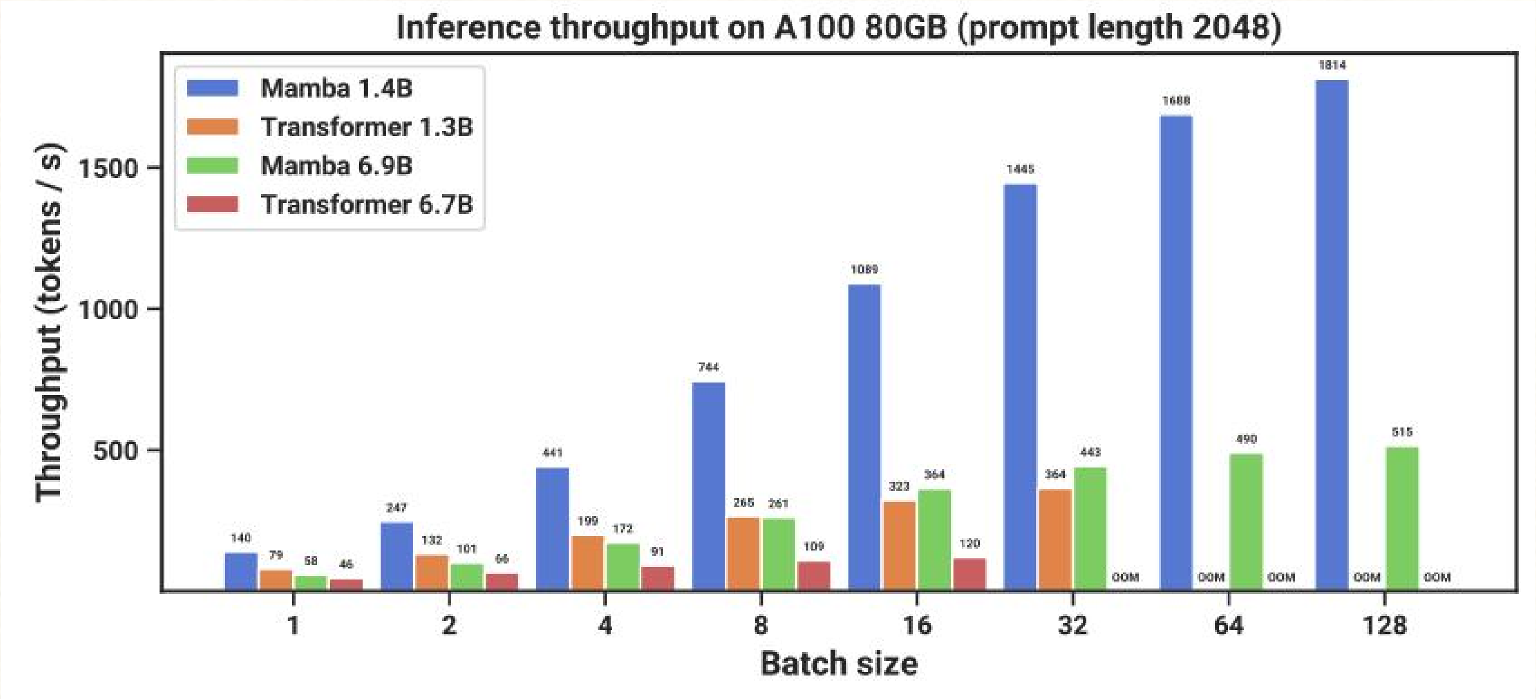


Qwen-next → 75% block
替换为**gated DeltaNet**

推理复杂度直接降到 O(n)

https://arxiv.org/abs/2406.06484
https://huggingface.co/Qwen/Qwen3-Next-80B-A3B-Instruct "Qwen/Qwen3-Next-80B-A3B-Instruct · Hugging Face"

Transformer 为什么难以继续扩展？

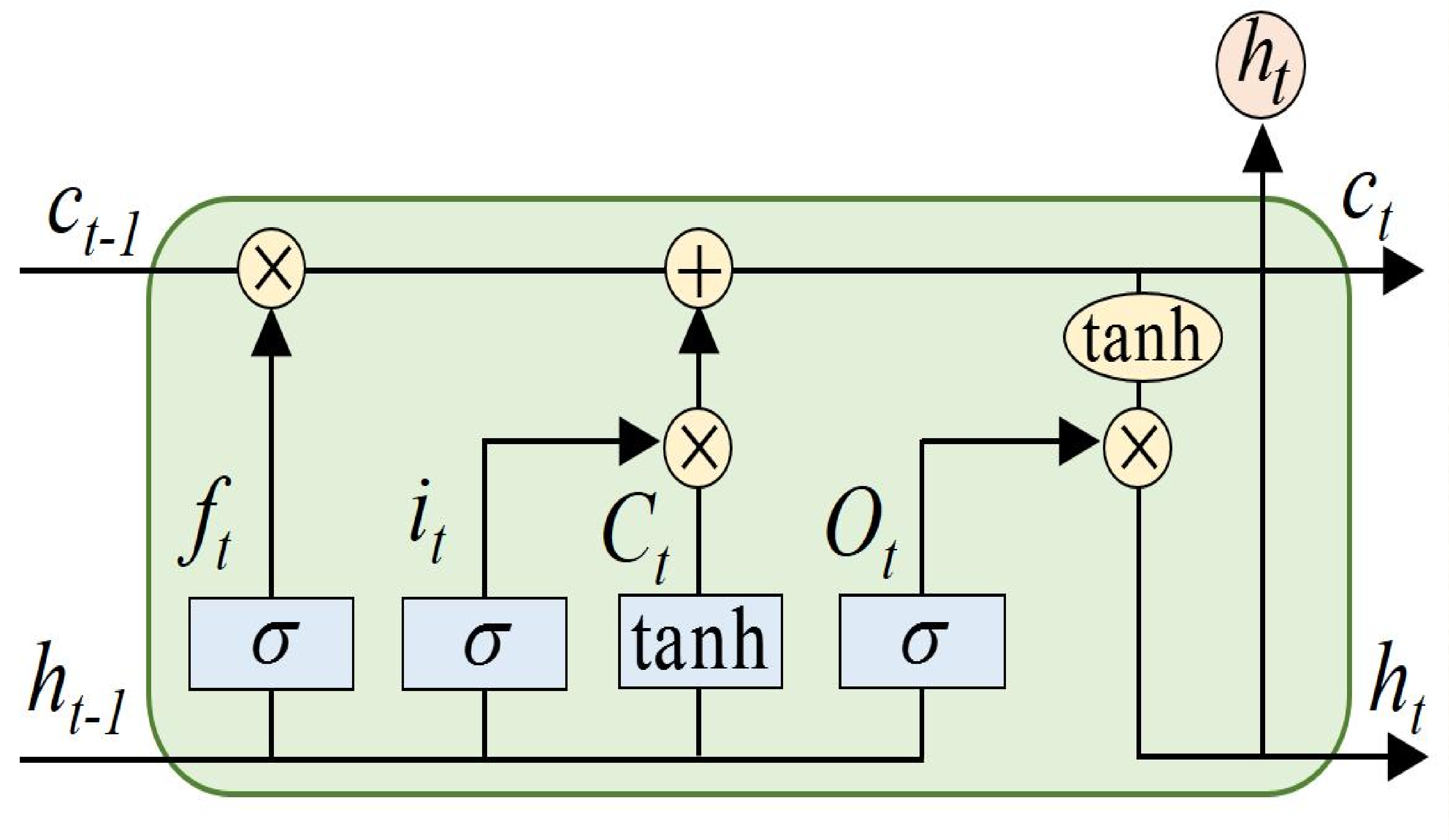


长序列 → O(N²) 的显存墙，单卡 A100 在 2 k 长度下就触顶

很多任务（语音、视频、RAG、Agent）需要长序列 + 高吞吐

https://arxiv.org/abs/2312.00752

回到起点：RNN/LSTM 的优势与局限



LSTM原文：https://www.bioinf.jku.at/publications/older/2604.pdf

RNN有没有平行训练的可能性

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$$S_t = f_{A,t}(S_{t-1}) + f_{B,t}(\boldsymbol{x_t})$$

$$\boldsymbol{y_t} = f_{C,t}(S_t)$$

$$f_{A,1}(S_0) = 0$$

$$\left\{\begin{array}{ll}
S_1 = f_{A,1}(S_0) + f_{B,1}(\boldsymbol{x_1}) & = f_{B,1}(\boldsymbol{x_1}) \\
S_2 = f_{A,2}(S_1) + f_{B,2}(\boldsymbol{x_2}) & = f_{A,2}\left(f_{B,1}(\boldsymbol{x_1})\right) + f_{B,2}(\boldsymbol{x_2}) \\
S_3 = f_{A,3}(S_2) + f_{B,3}(\boldsymbol{x_3}) & = f_{A,3}\left(f_{A,2}\left(f_{B,1}(\boldsymbol{x_1})\right) + f_{B,2}(\boldsymbol{x_2})\right) + f_{B,3}(\boldsymbol{x_3}) \\
\vdots & \\
S_t = f_{A,t}(S_{t-1}) + f_{B,t}(\boldsymbol{x_t}) & = \underbrace{f_{A,t}(f_{A,t-1} \ldots f_{A,3}(f_{A,2}}(f_{B,1}(\boldsymbol{x_1})\ldots) \quad \ldots + f_{B,t}(\boldsymbol{x_t})
\end{array}\right.$$

RNN有没有平行训练的可能性

$$S_t = S_{t-1} + f_{B,t}(\boldsymbol{x_t})$$

$$\boldsymbol{y_t} = f_{C,t}(S_t)$$

S_t is a $d \times d$ matric

$$f_{B,t}(\boldsymbol{x_t}) = D_t$$

$$f_{A,1}(S_0) = O$$

$$\begin{cases} S_1 = S_0 + f_{B,1}(\boldsymbol{x_1}) & = f_{B,1}(\boldsymbol{x_1}) \\ S_2 = S_1 + f_{B,2}(\boldsymbol{x_2}) & = f_{B,1}(\boldsymbol{x_1}) + f_{B,2}(\boldsymbol{x_2}) \\ S_3 = S_2 + f_{B,3}(\boldsymbol{x_3}) & = f_{B,1}(\boldsymbol{x_1}) + f_{B,2}(\boldsymbol{x_2}) + f_{B,3}(\boldsymbol{x_3}) \\ \vdots & \\ S_t = S_{t-1} + f_{B,t}(\boldsymbol{x_t}) & = f_{B,1}(\boldsymbol{x_1}) + f_{B,2}(\boldsymbol{x_2}) + f_{B,3}(\boldsymbol{x_3}) \quad \ldots\ldots + f_{B,t}(\boldsymbol{x_t}) \end{cases}$$

RNN有没有平行训练的可能性

$$S_t = S_{t-1} + f_{B,t}(\boldsymbol{x_t})$$

$$\boldsymbol{y_t} = f_{C,t}(S_t)$$

$$f_{A,1}(S_0) = O$$

$$\begin{cases} S_1 = D_1 & \boldsymbol{y_1} = D_1\boldsymbol{q_1} \\ S_2 = D_1 + D_2 & \boldsymbol{y_2} = D_1\boldsymbol{q_2} + D_2\boldsymbol{q_2} \\ S_3 = D_1 + D_2 + D_3 & \boldsymbol{y_3} = D_1\boldsymbol{q_3} + D_2\boldsymbol{q_3} + D_3\boldsymbol{q_3} \\ \vdots & \\ S_t = D_1 + D_2 + \ldots + D_t & \boldsymbol{y_t} = D_1\boldsymbol{q_t} + D_2\boldsymbol{q_t} + \ldots + D_t\boldsymbol{q_t} \end{cases}$$

S_t is a $d \times d$ matric

$$f_{B,t}(\boldsymbol{x_t}) = D_t$$

$$f_{C,t}(S_t) = S_t\boldsymbol{q_t}$$

$$\boldsymbol{q_t} = W_Q\boldsymbol{x_t}$$

RNN有没有平行训练的可能性

$$f_{A,1}(S_0) = O$$

$$\begin{cases} \boldsymbol{y_1} = D_1\boldsymbol{q_1} \\ \boldsymbol{y_2} = D_1\boldsymbol{q_2} + D_2\boldsymbol{q_2} \\ \boldsymbol{y_3} = D_1\boldsymbol{q_3} + D_2\boldsymbol{q_3} + D_3\boldsymbol{q_3} \\ \vdots \\ \boldsymbol{y_t} = D_1\boldsymbol{q_t} + D_2\boldsymbol{q_t} + \ldots + D_t\boldsymbol{q_t} \end{cases}$$

$$S_t = S_{t-1} + f_{B,t}(\boldsymbol{x_t})$$

$$\boldsymbol{y_t} = f_{C,t}(S_t)$$

S_t is a $d \times d$ matric

$$f_{B,t}(\boldsymbol{x_t}) = D_t$$

$$D_t = \boldsymbol{v_t}\boldsymbol{k_t}^T \qquad \boldsymbol{v_t} = W_v\boldsymbol{x_t} \qquad \boldsymbol{k_t} = W_k\boldsymbol{x_t}$$

$$f_{C,t}(S_t) = S_t\boldsymbol{q_t}$$

$$\boldsymbol{q_t} = W_Q\boldsymbol{x_t}$$

https://arxiv.org/abs/2006.16236

RNN有没有平行训练的可能性

$$f_{A,1}(S_0) = O$$

$$\begin{cases} \boldsymbol{y}_1 = \boldsymbol{v}_1 \boldsymbol{k}_1{}^T \boldsymbol{q}_1 \\ \boldsymbol{y}_2 = \boldsymbol{v}_1 \boldsymbol{k}_1{}^T \boldsymbol{q}_2 + \boldsymbol{v}_2 \boldsymbol{k}_2{}^T \boldsymbol{q}_2 \\ \boldsymbol{y}_3 = \boldsymbol{v}_1 \boldsymbol{k}_1{}^T \boldsymbol{q}_3 + \boldsymbol{v}_2 \boldsymbol{k}_2{}^T \boldsymbol{q}_3 + \boldsymbol{v}_3 \boldsymbol{k}_3{}^T \boldsymbol{q}_3 \\ \vdots \\ \boldsymbol{y}_t = \boldsymbol{v}_1 \boldsymbol{k}_1{}^T \boldsymbol{q}_t + \boldsymbol{v}_2 \boldsymbol{k}_2{}^T \boldsymbol{q}_t + \ldots + \boldsymbol{v}_t \boldsymbol{k}_t{}^T \boldsymbol{q}_t \end{cases}$$

$$S_t = S_{t-1} + f_{B,t}(\boldsymbol{x}_t)$$

$$\boldsymbol{y}_t = f_{C,t}(\mathrm{H}_t)$$

S_t is a $d \times d$ matric

$$f_{B,t}(\boldsymbol{x}_t) = D_t$$

$$D_t = \boldsymbol{v}_t \boldsymbol{k}_t{}^T \quad \boldsymbol{v}_t = W_v \boldsymbol{x}_t \quad \boldsymbol{k}_t = W_k \boldsymbol{x}_t$$

$$f_{C,t}(S_t) = S_t \boldsymbol{q}_t$$

$$\boldsymbol{q}_t = W_Q \boldsymbol{x}_t$$

RNN有没有平行训练的可能性

$$f_{A,1}(S_0) = O$$

$$\boldsymbol{y_t} = \boldsymbol{v_1}\boldsymbol{k_1}^T\boldsymbol{q_t} + \boldsymbol{v_2}\boldsymbol{k_2}^T\boldsymbol{q_t} + \ldots + \boldsymbol{v_t}\boldsymbol{k_t}^T\boldsymbol{q_t}$$

$$= \boldsymbol{v_1}a_{t,1} + \boldsymbol{v_2}a_{t,2} + \ldots + \boldsymbol{v_t}a_{t,t}$$

$$= a_{t,1}\boldsymbol{v_1} + a_{t,2}\boldsymbol{v_2} + \ldots + a_{t,t}\boldsymbol{v_t}$$

相较于Self Attention少
了 softmax

$$S_t = S_{t-1} + f_{B,t}(\boldsymbol{x_t})$$

$$\boldsymbol{y_t} = f_{C,t}(S_t)$$

S_t is a $d \times d$ matric

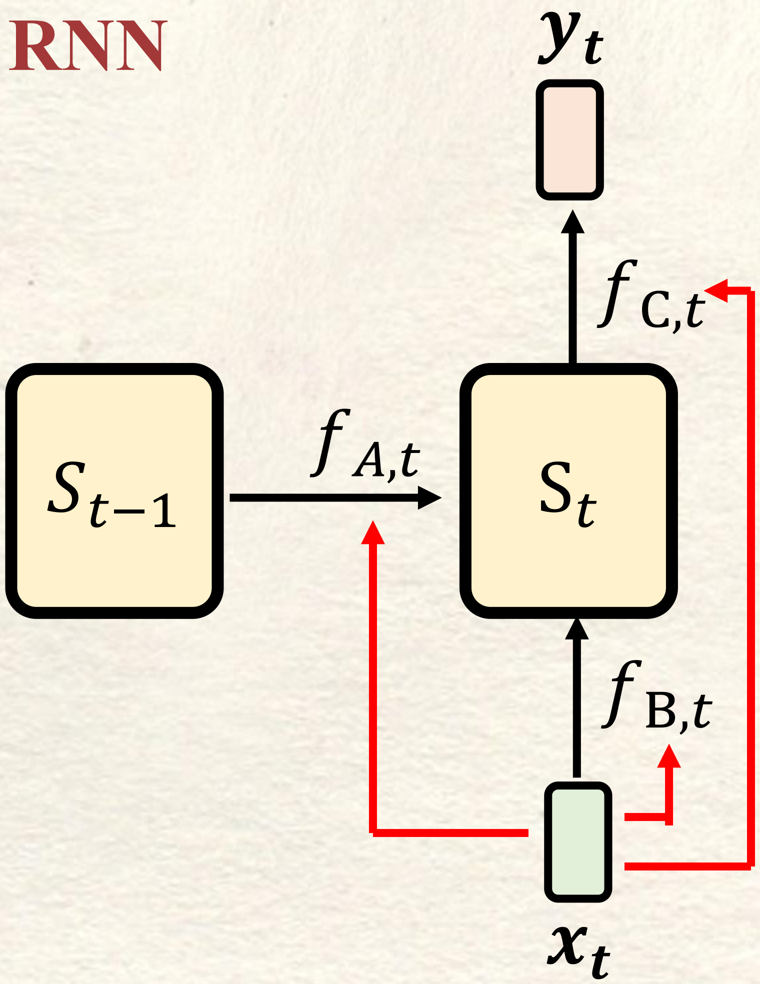
$$f_{B,t}(\boldsymbol{x_t}) = D_t$$

$$D_t = \boldsymbol{v_t}\boldsymbol{k_t}^T \qquad \boldsymbol{v_t} = W_v\boldsymbol{x_t} \qquad \boldsymbol{k_t} = W_k\boldsymbol{x_t}$$

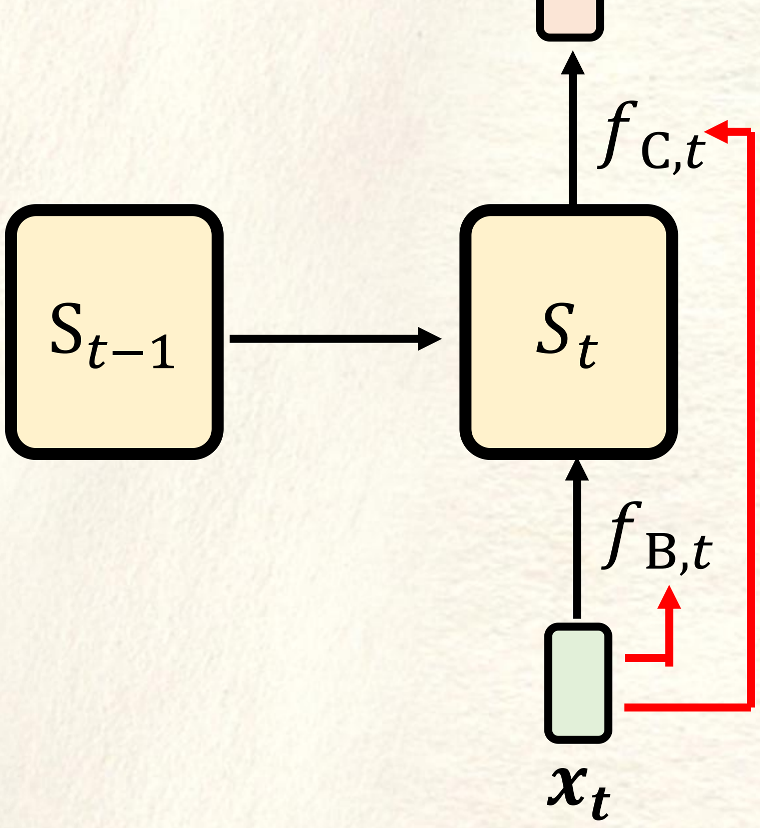
$$f_{C,t}(S_t) = S_t\boldsymbol{q_t}$$

$$\boldsymbol{q_t} = W_Q\boldsymbol{x_t}$$

RNN



Linear Attention



$$f_{C,t}(S_t) = S_t \boldsymbol{q_t}$$

$$\boldsymbol{q_t} = W_Q \boldsymbol{x_t}$$

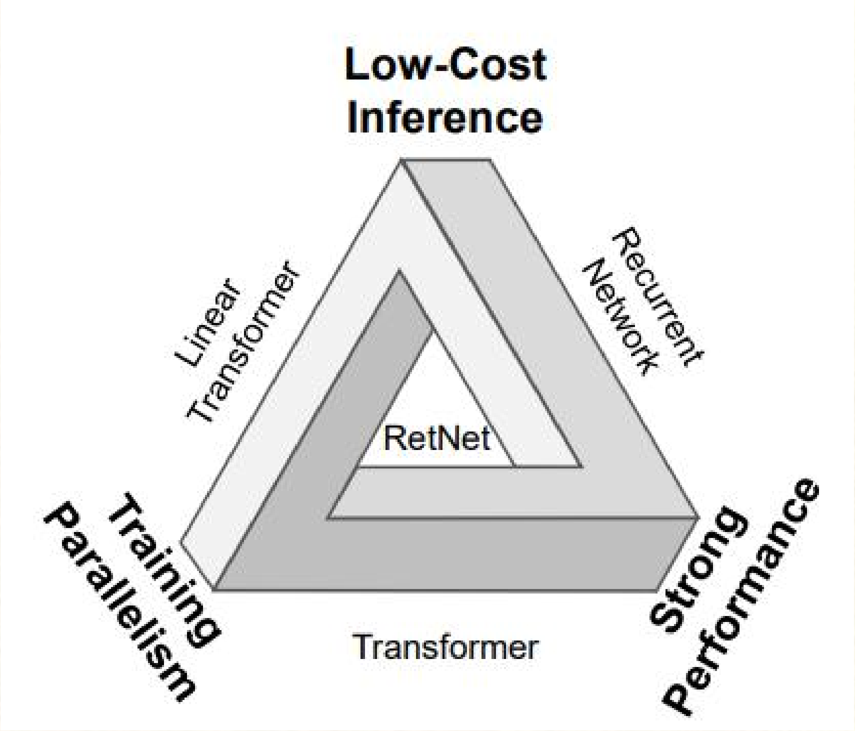
$$f_{B,t}(\boldsymbol{x_t}) = \boldsymbol{v_t} \boldsymbol{k_t}^T$$

$$\boldsymbol{v_t} = W_v \boldsymbol{x_t}$$

$$\boldsymbol{k_t} = W_k \boldsymbol{x_t}$$

Linear Attention 就是 **RNN 拿掉 “Reflection” $\mathbf{f_{A,t}}$**
因此在Inference阶段像 RNN，可以线性生成
Linear Attention 就是 **Self-attention 没有 Softmax**
因此在Training阶段像 RNN，可以并行训练

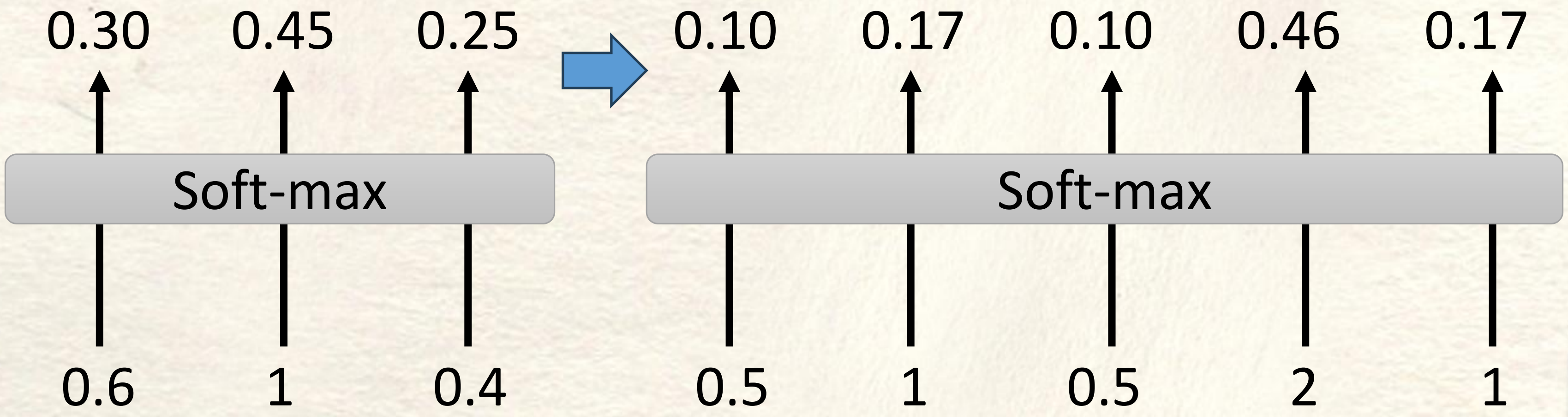
Linear Attention的局限



Linear attention提升推理速度是以牺牲记忆容量为代价的
这样会导致其表现效果不够理想

https://arxiv.org/abs/2307.08621

RetNet的洞察与改进



RetNet 指出 softmax 是 attention 的隐形漏斗

原本一骑绝尘的绝对高分，经它归一化后优势被摊薄，模型被迫“稀释”焦点，这种规律我们方便理解可以把它想象成——**选择性遗忘**

https://arxiv.org/abs/2307.08621

RetNet的洞察与改进

Linear Attention

$$S_t = S_{t-1} + \boldsymbol{v}_t \boldsymbol{k}_t^{T}$$

$$\boldsymbol{y}_t = S_t \boldsymbol{q}_t$$

RetNet

$$S_t = \gamma S_{t-1} + \boldsymbol{v}_t \boldsymbol{k}_t^{T}$$

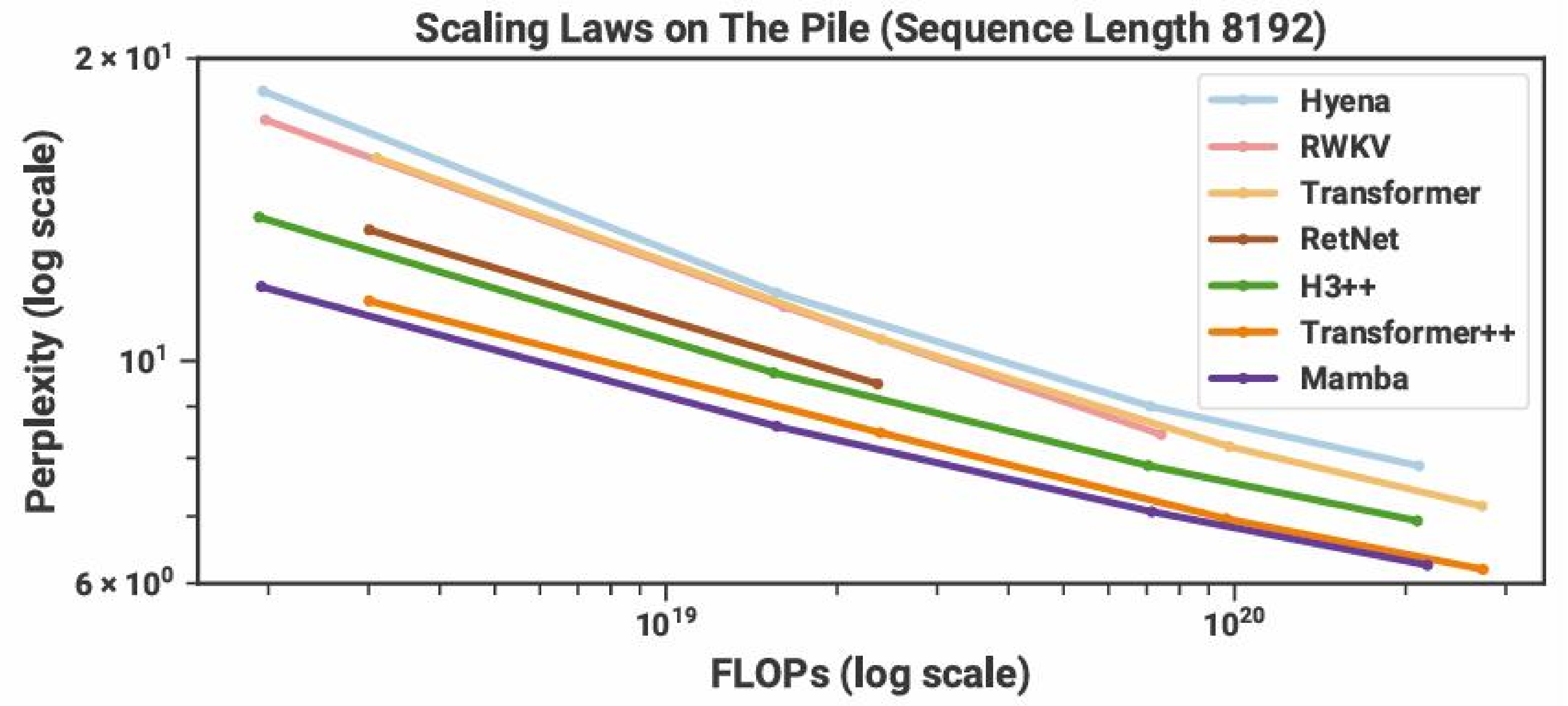
$$\boldsymbol{y}_t = S_t \boldsymbol{q}_t$$

RetNet引入MSR（Multi-Scale Retention，多尺度保留）

用“保留”代替“注意”，每头独享一个**指数衰减率** γ，在并行训练、循环推理、分块长序三种模式下，统一完成全局依赖建模。

https://arxiv.org/abs/2307.08621

Mamba是怎么火起来的？



https://arxiv.org/abs/2312.00752

YOCO和Mamba-2的改进

RetNet

$$\mathrm{S}_t = \gamma S_{t-1} + \boldsymbol{v}_t \boldsymbol{k}_t^{\ T}$$

$$\boldsymbol{y}_t = S_t \boldsymbol{q}_t$$

YOCO

$$\mathrm{S}_t = \gamma_t S_{t-1} + \boldsymbol{v}_t \boldsymbol{k}_t^{\ T}$$

$$\boldsymbol{y}_t = S_t \boldsymbol{q}_t$$

$$\gamma_t = sigmoid(\boldsymbol{v}_t)$$

YOCO和Mamba-2将固定的指数衰减率 γ优化为输入依赖的门控衰减。

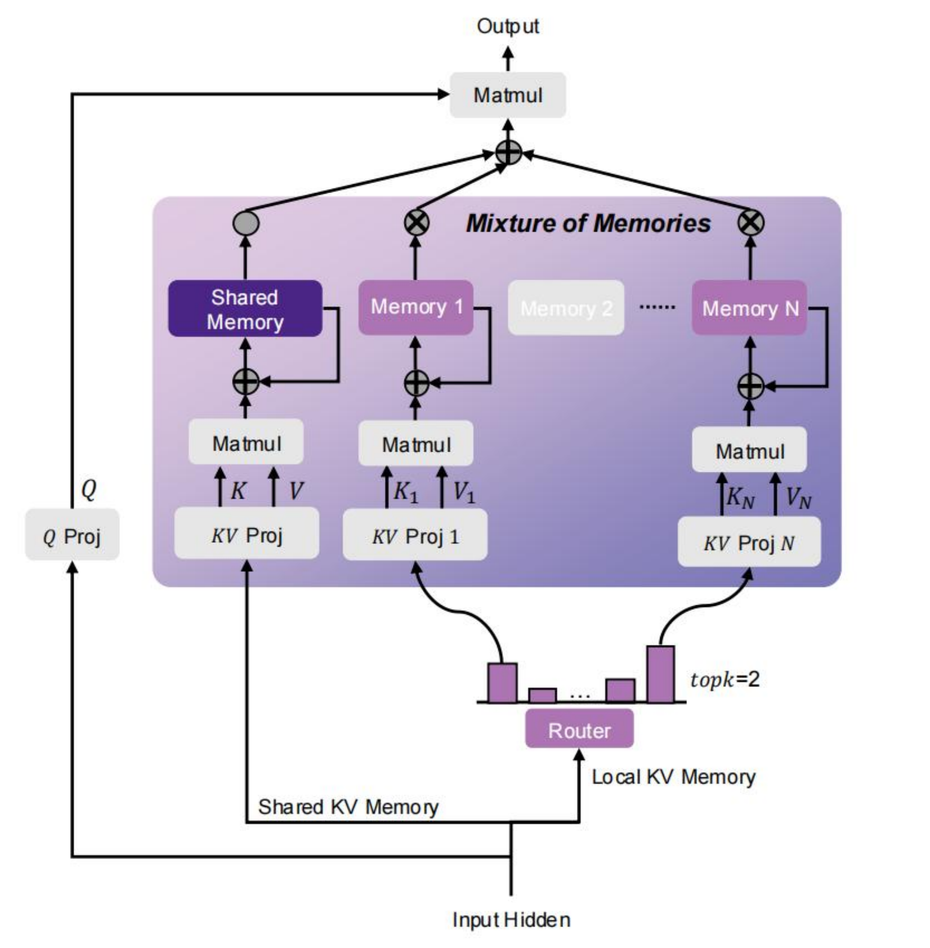
https://arxiv.org/abs/2405.05254

更多在状态转移方面的改进

Model	Recurrence	Memory read-out
Linear Attention [48, 47]	$\mathbf{S}_t = \mathbf{S}_{t-1} + \boldsymbol{v}_t \boldsymbol{k}_t^\top$	$\boldsymbol{o}_t = \mathbf{S}_t \boldsymbol{q}_t$
+ Kernel	$\mathbf{S}_t = \mathbf{S}_{t-1} + \boldsymbol{v}_t \phi(\boldsymbol{k}_t)^\top$	$\boldsymbol{o}_t = \mathbf{S}_t \phi(\boldsymbol{q}_t)$
+ Normalization	$\mathbf{S}_t = \mathbf{S}_{t-1} + \boldsymbol{v}_t \phi(\boldsymbol{k}_t)^\top, \quad \boldsymbol{z}_t = \boldsymbol{z}_{t-1} + \phi(\boldsymbol{k}_t)$	$\boldsymbol{o}_t = \mathbf{S}_t \phi(\boldsymbol{q}_t) / (\boldsymbol{z}_t^\top \phi(\boldsymbol{q}_t))$
DeltaNet [101]	$\mathbf{S}_t = \mathbf{S}_{t-1}(\mathbf{I} - \beta_t \boldsymbol{k}_t \boldsymbol{k}_t^\top) + \beta_t \boldsymbol{v}_t \boldsymbol{k}_t^\top$	$\boldsymbol{o}_t = \mathbf{S}_t \boldsymbol{q}_t$
Gated RFA [81]	$\mathbf{S}_t = g_t \mathbf{S}_{t-1} + (1 - g_t) \boldsymbol{v}_t \boldsymbol{k}_t^\top, \quad \boldsymbol{z}_t = g_t \boldsymbol{z}_{t-1} + (1 - g_t) \boldsymbol{k}_t$	$\boldsymbol{o}_t = \mathbf{S}_t \boldsymbol{q}_t / (\boldsymbol{z}_t^\top \boldsymbol{q}_t)$
S4 [32, 106]	$\mathbf{S}_t = \mathbf{S}_{t-1} \odot \exp(-(\boldsymbol{\alpha} \mathbf{1}^\top) \odot \exp(\boldsymbol{A})) + \boldsymbol{B} \odot (\boldsymbol{v}_t \mathbf{1}^\top)$	$\boldsymbol{o}_t = (\mathbf{S}_t \odot \boldsymbol{C}) \mathbf{1} + \boldsymbol{d} \odot \boldsymbol{v}_t$
ABC [82]	$\mathbf{S}_t^k = \mathbf{S}_{t-1}^k + \boldsymbol{k}_t \boldsymbol{\phi}_t^\top, \quad \mathbf{S}_t^v = \mathbf{S}_{t-1}^v + \boldsymbol{v}_t \boldsymbol{\phi}_t^\top$	$\boldsymbol{o}_t = \mathbf{S}_t^v \,\mathrm{softmax}(\mathbf{S}_t^k \boldsymbol{q}_t)$
DFW [63]	$\mathbf{S}_t = \mathbf{S}_{t-1} \odot (\boldsymbol{\beta}_t \boldsymbol{\alpha}_t^\top) + \boldsymbol{v}_t \boldsymbol{k}_t^\top$	$\boldsymbol{o}_t = \mathbf{S}_t \boldsymbol{q}_t$
RetNet [108]	$\mathbf{S}_t = \gamma \mathbf{S}_{t-1} + \boldsymbol{v}_t \boldsymbol{k}_t^\top$	$\boldsymbol{o}_t = \mathbf{S}_t \boldsymbol{q}_t$
Mamba [31]	$\mathbf{S}_t = \mathbf{S}_{t-1} \odot \exp(-(\boldsymbol{\alpha}_t \mathbf{1}^\top) \odot \exp(\boldsymbol{A})) + (\boldsymbol{\alpha}_t \odot \boldsymbol{v}_t) \boldsymbol{k}_t^\top$	$\boldsymbol{o}_t = \mathbf{S}_t \boldsymbol{q}_t + \boldsymbol{d} \odot \boldsymbol{v}_t$
GLA [124]	$\mathbf{S}_t = \mathbf{S}_{t-1} \odot (\mathbf{1} \boldsymbol{\alpha}_t^\top) + \boldsymbol{v}_t \boldsymbol{k}_t^\top = \mathbf{S}_{t-1} \mathrm{Diag}(\boldsymbol{\alpha}_t) + \boldsymbol{v}_t \boldsymbol{k}_t^\top$	$\boldsymbol{o}_t = \mathbf{S}_t \boldsymbol{q}_t$
RWKV-6 [79]	$\mathbf{S}_t = \mathbf{S}_{t-1} \mathrm{Diag}(\boldsymbol{\alpha}_t) + \boldsymbol{v}_t \boldsymbol{k}_t^\top$	$\boldsymbol{o}_t = (\mathbf{S}_{t-1} + (\boldsymbol{d} \odot \boldsymbol{v}_t) \boldsymbol{k}_t^\top) \boldsymbol{q}_t$
HGRN-2 [92]	$\mathbf{S}_t = \mathbf{S}_{t-1} \mathrm{Diag}(\boldsymbol{\alpha}_t) + \boldsymbol{v}_t (\mathbf{1} - \boldsymbol{\alpha}_t)^\top$	$\boldsymbol{o}_t = \mathbf{S}_t \boldsymbol{q}_t$
mLSTM [9]	$\mathbf{S}_t = f_t \mathbf{S}_{t-1} + i_t \boldsymbol{v}_t \boldsymbol{k}_t^\top, \quad \boldsymbol{z}_t = f_t \boldsymbol{z}_{t-1} + i_t \boldsymbol{k}_t$	$\boldsymbol{o}_t = \mathbf{S}_t \boldsymbol{q}_t / \max\{1, \lvert \boldsymbol{z}_t^\top \boldsymbol{q}_t \rvert\}$
Mamba-2 [19]	$\mathbf{S}_t = \gamma_t \mathbf{S}_{t-1} + \boldsymbol{v}_t \boldsymbol{k}_t^\top$	$\boldsymbol{o}_t = \mathbf{S}_t \boldsymbol{q}_t$
GSA [131]	$\mathbf{S}_t^k = \mathbf{S}_{t-1}^k \mathrm{Diag}(\boldsymbol{\alpha}_t) + \boldsymbol{k}_t \boldsymbol{\phi}_t^\top, \quad \mathbf{S}_t^v = \mathbf{S}_{t-1}^v \mathrm{Diag}(\boldsymbol{\alpha}_t) + \boldsymbol{v}_t \boldsymbol{\phi}_t^\top$	$\boldsymbol{o}_t = \mathbf{S}_t^v \,\mathrm{softmax}(\mathbf{S}_t^k \boldsymbol{q}_t)$
Gated DeltaNet [125]	$\mathbf{S}_t = \mathbf{S}_{t-1} \left(\alpha_t (\mathbf{I} - \beta_t \boldsymbol{k}_t \boldsymbol{k}_t^\top)\right) + \beta_t \boldsymbol{v}_t \boldsymbol{k}_t^\top$	$\boldsymbol{o}_t = \mathbf{S}_t \boldsymbol{q}_t$

https://arxiv.org/abs/2406.06484

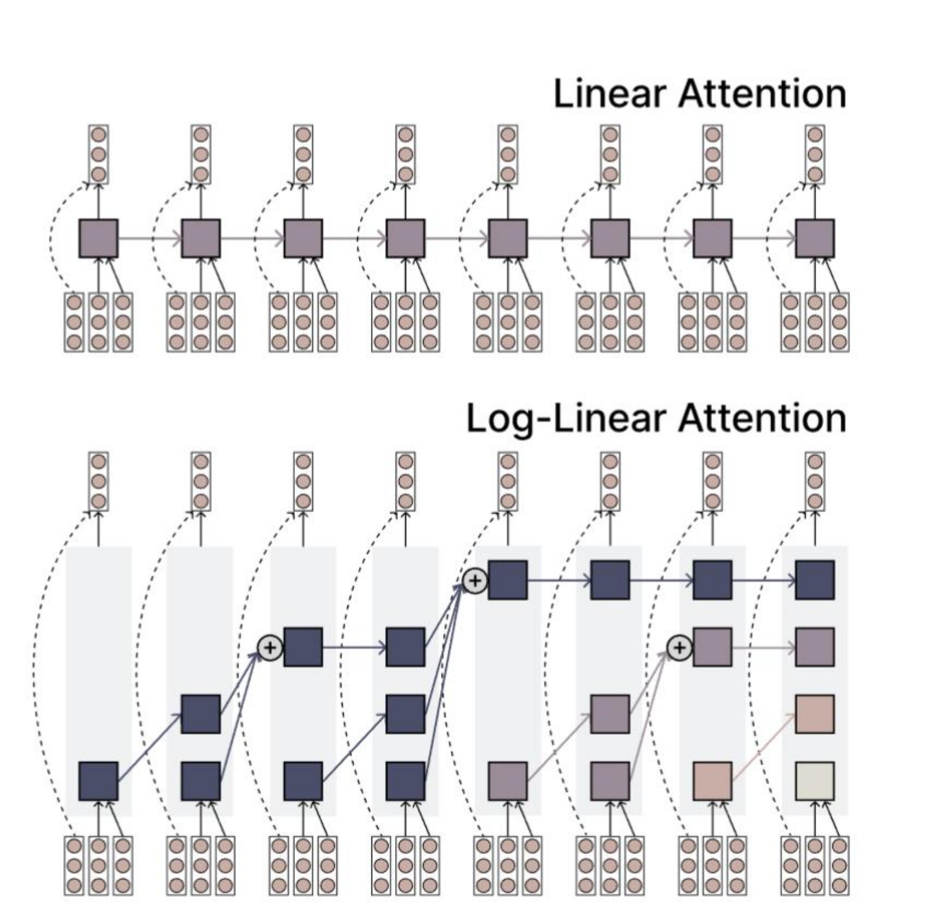
Linear Attention的最新进展



MoM 提出“多记忆混合”机制，用独立 RNN 记忆槽+稀疏路由替代单一隐藏态，在线性训练和常数推理复杂度下显著扩大容量、抑制干扰，使线性注意力模型在多项召回密集型任务上首次逼近甚至超越 Transformer。

https://arxiv.org/abs/2502.13685

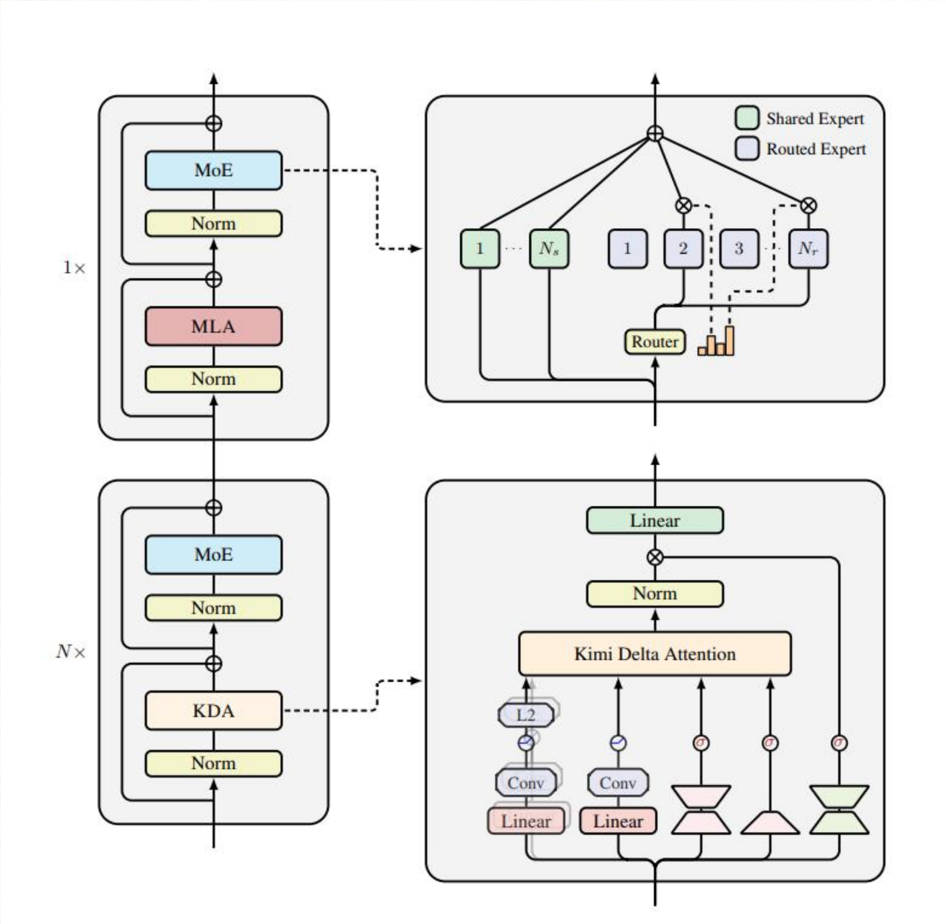
Linear Attention的最新进展



Log-Linear Attention 用随序列长度对数增长的隐状态取代线性注意力固定的隐状态，实现训练复杂度 O(T log T)、推理内存 O(log T)，在保持高效并行的同时显著提升长程关联建模能力。

https://arxiv.org/abs/2506.04761

Linear Attention的最新进展



Kimi Linear 提出的高效混合线性注意力架构，在降低75% KV缓存与6倍解码速度的同时，性能全面超越全注意力基线。

https://arxiv.org/abs/2510.26692

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感谢各位老师同学批评与指导

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